4.10 Explain how data from seismometers can be used to identify the location of an earthquake

4.11 **Demonstrate an understanding of how P and S waves travel inside the Earth including reflection and refraction**

4.12 Explain how the Earth's outermost layer is composed of (tectonic) plates and is in relative motion due to convection currents in the mantle

4.13 Demonstrate an understanding of how, at plate boundaries, plates may slide past each other, sometimes causing earthquakes

Topic 5

Generation and transmission of electricity

5.1 Describe current as the rate of flow of charge and voltage as an electrical pressure giving a measure of the energy transferred

5.2 Define power as the energy transferred per second and measured in watts

5.3 Use the equation:

electrical power (watt, W) = current (ampere, A) × potential difference (volt, V)
$P = I \times V$

5.4 *Investigate the power consumption of low-voltage electrical items*

5.5 Discuss the advantages and disadvantages of methods of large-scale electricity production using a variety of renewable and non-renewable resources

5.6 Demonstrate an understanding of the factors that affect the size and direction of the induced current

5.7 *Investigate factors affecting the generation of electric current by induction*

5.8 Explain how to produce an electric current by the relative movement of a magnet and a coil of wire:

- a on a small scale
- b in the large-scale generation of electrical energy

5.9 Recall that generators supply current which alternates in direction

5.10 Explain the difference between direct and alternating current